F79MB Coursework: Assessed Project 2

1. a) Pairwise relationships and correlation coefficients, r between variables `speed`, `signal`, `bandwidth` and `legacy_users` are shown in the figure below.

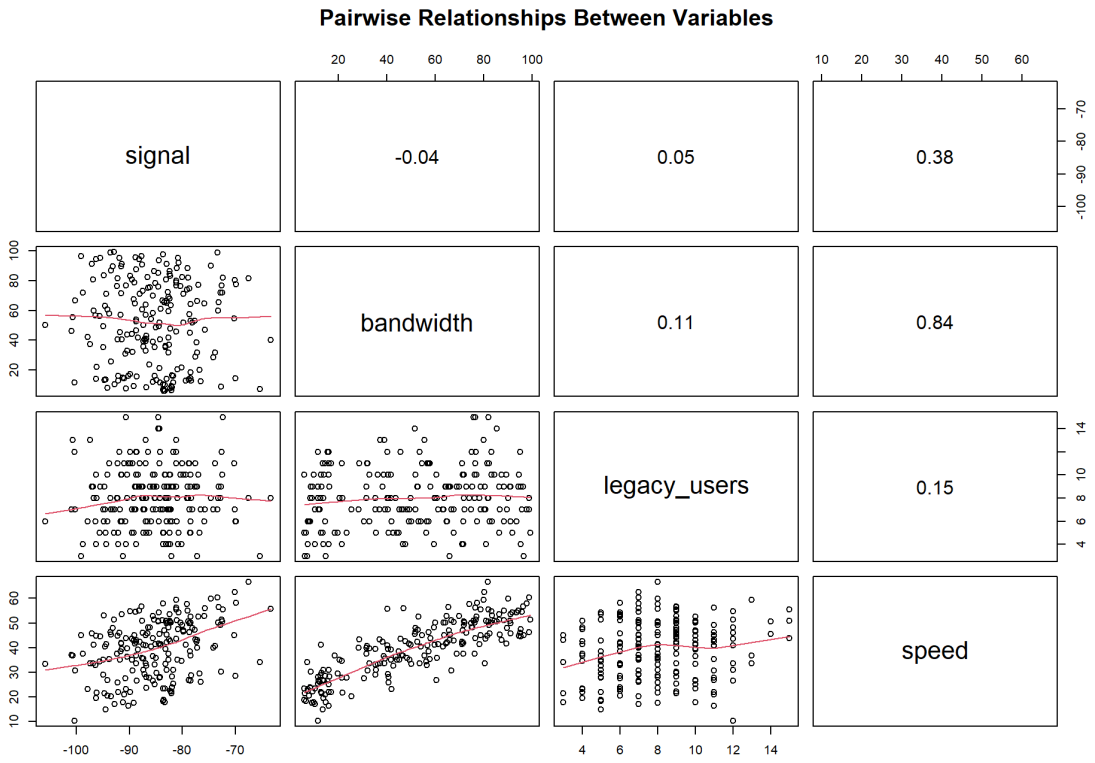


Figure 1: Pairwise relationships and correlation coefficients, r between variables extracted from a sample of 200 network observations in Heriot-Watt University

Figure 1 presents a pairs plot showing the pairwise relationships and corresponding correlation coefficients between the variables. From the figure, it can be seen that the response variable `speed` and the remaining explanatory variables are positively correlated, with `bandwidth` achieving the most positive correlation at $r = 0.84$, followed by `signal` at $r = 0.38$ and finally `legacy_users` at $r = 0.15$. This indicates that larger bandwidth and signal is associated with higher speed. Hence, both `bandwidth` and `signal` are likely to be an important predictor in a model for speed. On the other hand, the relatively weak correlation between `legacy_users` and `speed` suggests that it may not be a strong linear predictor of speed, although it could still contribute in a multivariate setting.

It is also worth noting that comparing between explanatory variables yields correlation coefficients that are close to 0. This signifies little multicollinearity issues between the explanatory variables. Moreover, the curved smooth lines also suggest that some relationships, especially involving `speed`, may not be perfectly linear, so it may be worth considering a nonlinear term or interaction in the model.

b) Fitting the proposed model (**Model 1**), which includes `speed` as the response variable modeled with explanatory variables `bandwidth`, `signal` and `legacy_users`, excluding interaction terms,

we obtain the below summary output results:

```
lm(formula = speed ~ bandwidth + signal + legacy_users)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  74.53417    3.62549  20.558   <2e-16 ***
bandwidth     0.34899    0.01068  32.681   <2e-16 ***
signal        0.63770    0.04021  15.861   <2e-16 ***
legacy_users  0.15153    0.11669   1.299    0.196

Residual standard error: 4.215 on 196 degrees of freedom
Multiple R-squared:   0.87,     Adjusted R-squared:  0.868
F-statistic: 437.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

The F-test indicates very strong evidence ($p < 2.2 \times 10^{-16}$) that this model fits better than the null model. This indicates that at least one of the explanatory variables provides significant information about mobile data speed. We also notice the R^2 value being 0.87, meaning that **Model 1** explains 87% of the variability in the mobile data speed, with the remaining 13% of the variation unexplained by the model. While these are two good indicators, we move on to determining the significance of the predictors. Significant predictors include `bandwidth` ($p < 2 \times 10^{-16}$) and `signal` ($p < 2 \times 10^{-16}$). Meanwhile, `legacy_users` ($p = 0.196$) do not significantly contribute to the model at the 5% significance level. Thus, we can simplify the model by removing this term.

It should be noted that these analysis of variance (ANOVA) results are sequential and depend on the order in which variables are entered into the model. Hence, we arrange the explanatory variables in the order that ensures Sum of Squares monotonically decreases for every additional explanatory variable added into the model. Thus, we consider the ANOVA results for **Model 1**, as shown below:

```
              Df Sum Sq Mean Sq  F value Pr(>F)
bandwidth      1  18748   18748 1055.326 <2e-16 ***
signal         1   4526    4526  254.754 <2e-16 ***
legacy_users   1     30      30    1.686  0.196
Residuals    196   3482      18
```

We see that `bandwidth` is highly significant ($p < 2 \times 10^{-16}$). With `bandwidth` included, `signal` is also highly significant ($p < 2 \times 10^{-16}$). However, note that with the previous 2 main effects included in the model, `legacy_users` is not significant ($p = 0.196$). Therefore, we can simplify the model by removing `legacy_users`.

Now fit an additive **Model 2** (with both main effects, but without `legacy_users`):

```
lm(formula = speed ~ bandwidth + signal)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 75.92919    3.46870   21.89   <2e-16 ***
bandwidth    0.35055    0.01063   32.98   <2e-16 ***
signal       0.64067    0.04021   15.93   <2e-16 ***

Residual standard error: 4.222 on 197 degrees of freedom
Multiple R-squared:  0.8689,    Adjusted R-squared:  0.8676
F-statistic: 652.8 on 2 and 197 DF,  p-value: < 2.2e-16
```

Compared to the full model, R^2 has only decreased slightly (0.11%) to explaining 86.89% of the variance in the data. Furthermore, the adjusted R^2 only decreased by 0.04%. The F-test remains highly significant ($p < 2 \times 10^{-16}$). This confirms that this simpler model (**Model 2**) should be preferred, and we can drop the variable `legacy_users` since all individual coefficients are now significant (p-values are all less than 2×10^{-16}).

We further investigate the ANOVA table of **Model 2**:

```
            Df Sum Sq Mean Sq F value Pr(>F)
bandwidth    1  18748   18748  1051.7 <2e-16 ***
signal       1   4526    4526   253.9 <2e-16 ***
Residuals  197   3512      18
```

We observe that `bandwidth` ($p < 2 \times 10^{-16}$) and `signal` ($p < 2 \times 10^{-16}$) are both highly significant, highlighting their importance as predictors in explaining the variability of the mobile data speed in **Model 2**.

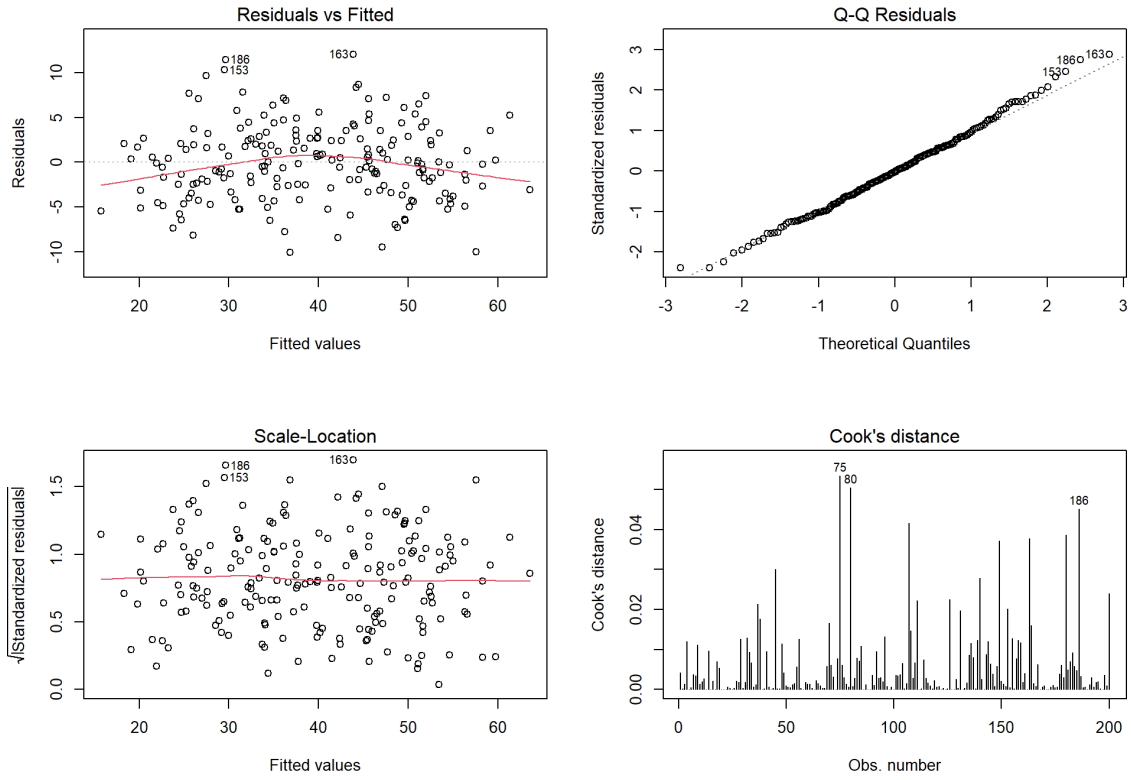


Figure 2: Residuals plots of **Model 2**

Upon examination of the residuals of **Model 2** (see Figure 2), the plot of residuals versus fitted values shows potential non-linearity, with high possibility of a quadratic pattern. The normal QQ plot suggests a slightly right-skewed distribution of residuals, with a subtle upward curve above the straight line at the upper tail of the plot. However, this deviation can be considered minor, with no significant deviations from the normality assumption. The scale-location plot suggests that the model has homoscedasticity, with variance remaining constant throughout. The Cook's distance plot values show no overly influential points, as all points are well below the threshold of 1. Hence, there is no need to exclude any points for model re-fitting.

Thus, excluding interaction terms, **Model 2** is preferred and its fitted model is:

$$\text{Speed}_i = 75.92919 + 0.35055(\text{bandwidth}_i) + 0.64067(\text{signal}_i)$$

Interpretation: The preferred **Model 2** suggests that mobile data speed increases by 0.35055 for every 1 MHz increase in the available network bandwidth, holding the mobile signal strength constant. Also, mobile data speed increases by 0.64067 for every 1 dBm increase in mobile signal strength, holding the available network bandwidth constant. It is worth noting that signal values are negative, hence, every 1 dBm increase in mobile signal strength decreases the reduction of mobile data speed, again holding the available network bandwidth constant.

Conclusion: The preferred **Model 2**, which includes the explanatory variables `bandwidth` and `signal`, provides valuable insights into mobile data speed, with both the F-statistic and all coefficient t-tests being highly significant. The fitted model suggests that as available network bandwidth and mobile signal strength increases, the mobile data speed also increases. With an R^2 value of 0.8689, the model explains approximately 86.89% of the variability in mobile data speed based on these explanatory variables. This result is satisfactory. However, there are concerns with the model diagnostics. The main concern is due to the "Residuals vs Fitted" plot suggesting a slight non-linear, possibly quadratic pattern. Therefore, the linearity assumption has been imperfectly met for now. In conclusion, while **Model 2** provides satisfactory insights, there is still room for further refinement.

Discussion of further analyses for refining the model:

(a) **Non-linear modeling:** The "Residuals vs Fitted" plot suggests a potential non-linear, possibly quadratic pattern. Hence, a more complex model may be required to achieve the linearity assumption and better capture the relationship between these variables. Some approaches to address this includes adding polynomial terms, adding interaction terms, exploring non-linear regression models, and many other techniques.

(b) **Use of Alternative Models:** The current model assumes normality, homoscedasticity and linearity, which may not fully hold. To improve the model's performance, alternative modelling techniques could be considered such as spline regression, generalized least squares (GLS), generalized linear models (GLM) and various machine learning methods could be considered.

(c) **Transformation of variables:** The small issue with verifying the normality assumption in the residuals suggest that transformations of certain variables may be helpful. For instance, applying a log transform to the response variable `speed` or the `bandwidth` predictor could help stabilize variance and help to meet normality assumptions.

(d) **Adding New Variables:** Adding a categorical variable for the "Network Type" (e.g., 4G vs. 5G) would likely explain a large portion of the remaining 13% of unexplained variability.

By exploring non-linear relationships, applying transformations to variables, considering the use of alternative models and adding more variables, the model can be further improved to better explain mobile data speed and increase predictive accuracy.

c) Fitting the proposed model (**Model 3**), which includes `speed` as the response variable and `bandwidth`, its quadratic term, `signal`, and the interaction between the quadratic term and `signal`, we obtain the below summary output results:

```
lm(formula = speed ~ bandwidth + signal * I(bandwidth^2))

Coefficients:
```

```
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             7.421e+01  5.269e+00  14.085   <2e-16 ***
bandwidth               5.481e-01  4.394e-02  12.473   <2e-16 ***
signal                  6.607e-01  6.198e-02  10.661   <2e-16 ***
I(bandwidth^2)         -2.455e-03  1.300e-03  -1.889   0.0604 .
signal:I(bandwidth^2) -5.822e-06  1.355e-05  -0.430   0.6679

Residual standard error: 4.028 on 195 degrees of freedom
Multiple R-squared:  0.8819,    Adjusted R-squared:  0.8795
F-statistic:   364 on 4 and 195 DF,  p-value: < 2.2e-16
```

The F-test indicates very strong evidence ($p < 2.2 \times 10^{-16}$) that this model fits better than the null model. This indicates that at least one of the explanatory variables provides significant information about mobile data speed. We also notice the R^2 value being 0.8819, meaning that **Model 3** explains 88.19% of the variability in the mobile data speed, with the remaining 11.81% of the variation unexplained by the model. While these are good indicators, we move on to determining the significance of the predictors. Significant predictors include `bandwidth` ($p < 2 \times 10^{-16}$) and `signal` ($p < 2 \times 10^{-16}$). Meanwhile, the quadratic term of `bandwidth` ($p = 0.0604$) and the interaction between the quadratic term and `signal` ($p = 0.6679$) do not significantly contribute to the model at the 5% significance level. However, it should be noted that the quadratic term of `bandwidth` ($p = 0.0604$) does significantly contribute to the model at the 10% significance level. Thus, more investigation is required before any removal of terms are decided.

It should be noted that these ANOVA results are sequential and depend on the order in which variables are entered into the model. Hence, we arrange the explanatory variables in the order that ensures Sum of Squares monotonically decreases for every additional explanatory variable added into the model. Thus, we consider the ANOVA results for Model 3, as shown below:

```
                       Df Sum Sq Mean Sq  F value   Pr(>F)
bandwidth               1  18748   18748 1155.567  < 2e-16 ***
signal                  1   4526    4526  278.952  < 2e-16 ***
I(bandwidth^2)          1    345     345   21.279 7.18e-06 ***
signal:I(bandwidth^2)   1      3       3    0.185    0.668
Residuals             195   3164      16
```

We see that `bandwidth` is highly significant ($p < 2 \times 10^{-16}$). With `bandwidth` included, `signal` is also highly significant ($p < 2 \times 10^{-16}$). With the previous 2 effects included, the quadratic term of `bandwidth` is also highly significant ($p < 7.18 \times 10^{-6}$). However, note that with the previous 3 main effects included in the model, the interaction between the quadratic term and `signal` is not significant ($p = 0.668$). It is worth noting that there is a discrepancy between the p-values obtained from the coefficient summary and the ANOVA table. The ANOVA table reports sequential (Type I) sums of squares, where the significance of the quadratic term is assessed after accounting for `bandwidth` and `signal`. In contrast, the coefficient p-values assess the contribution of each variable conditional on all other variables in the model. This explains why the quadratic term appears significant in the ANOVA table but marginally insignificant in the coefficient output. Therefore, we can simplify the model by removing the interaction between the quadratic term and `signal`.

Now fit an additive **Model 4** (with the interaction between the quadratic term and `signal` term removed):

```
lm(formula = speed ~ bandwidth + signal + I(bandwidth^2))
```

```
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    72.4761204  3.3856251  21.407  < 2e-16 ***
bandwidth       0.5449858  0.0432627  12.597  < 2e-16 ***
signal          0.6398131  0.0382801  16.714  < 2e-16 ***
I(bandwidth^2) -0.0019263  0.0004167  -4.623 6.87e-06 ***

Residual standard error: 4.02 on 196 degrees of freedom
Multiple R-squared:  0.8818,    Adjusted R-squared:   0.88
F-statistic: 487.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

Compared to the full model, R^2 has only decreased slightly (0.01%) to explain 88.18% of the variance in the data. Furthermore, the adjusted R^2 also increased by 0.05%. The F-test remains highly significant ($p < 2.2 \times 10^{-16}$). This confirms that this simpler model (**Model 4**) should be preferred, and we can drop the interaction between the quadratic term and `signal` since all individual coefficients are now significant (p-values are less than 2×10^{-16} for both `signal` and `bandwidth`, while `bandwidth`'s quadratic term has a p-value of 6.87×10^{-6}).

We further investigate the ANOVA table of **Model 4**:

```
                Df Sum Sq Mean Sq F value   Pr(>F)
bandwidth        1  18748   18748 1160.39  < 2e-16 ***
signal           1   4526    4526  280.12  < 2e-16 ***
I(bandwidth^2)   1    345     345   21.37 6.87e-06 ***
Residuals      196   3167      16
```

We observe that all predictors used highly significant (p-values are less than 2×10^{-16} for both `signal` and `bandwidth`, while `bandwidth`'s quadratic term has a p-value of 6.87×10^{-6}), highlighting their importance as predictors in explaining the variability of the mobile data speed in **Model 4**.

Upon examination of the residuals of **Model 4** (see Figure 3), the plot of residuals versus fitted values shows no particular pattern, though there appears to be one potential outlier at the observation number 186. The normal QQ plot is observed to have deviations at the upper and lower tails, suggesting the residuals have slight heavy-tails. The model is under-predicting and over-predicting slightly more than a normal distribution would expect for some cases. However, this deviation can be considered minor, with no significant deviations from the normality assumption. The scale-location plot suggests that the model has homoscedasticity, with variance remaining constant throughout. The Cook's distance plot values show no overly influential points, as all points are well below the threshold of 1. In particular, observation 186 does not seem overly influential, as its Cook's distance value is small (0.0567). Hence, there is no need to exclude any points for model re-fitting.

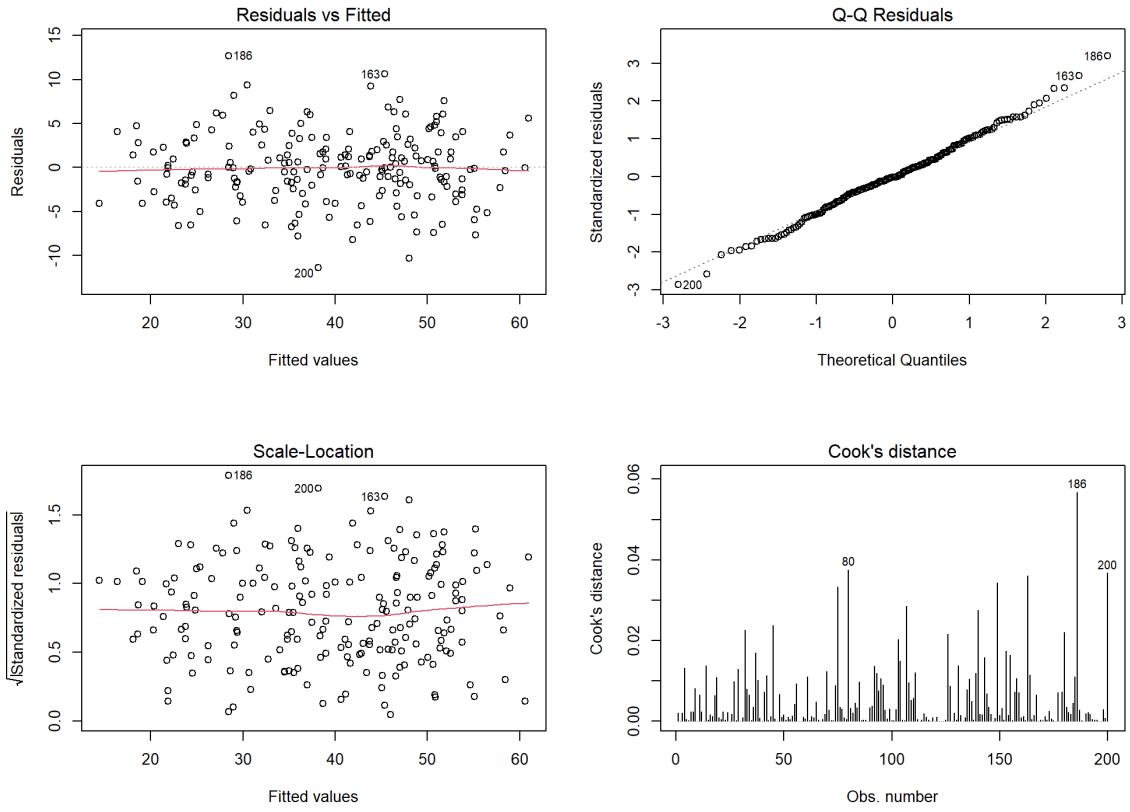


Figure 3: Residuals plots of **Model 4**

Thus, **Model 4** is preferred and the fitted model is:

$$\text{Speed}_i = 72.4761204 + 0.5449858(\text{bandwidth}_i) + 0.6398131(\text{signal}_i) - 0.0019263(\text{bandwidth}_i^2)$$

Interpretation: The preferred **Model 4** suggests that mobile data speed increases by 0.5449858 for every 1 MHz increase in the available network bandwidth, holding all other variables constant. Also, mobile data speed increases by 0.6398131 for every 1 dBm increase in mobile signal strength, holding all other variables constant. Furthermore, every 1 MHz increase in the quadratic term of the available network bandwidth decreases mobile data speed by 0.0019263, holding all other variables constant. It is worth noting that signal values are negative, hence, every 1 dBm increase in mobile signal strength decreases the reduction of mobile data speed, again holding all other variables constant.

Conclusion: The preferred **Model 4**, which includes the explanatory variables `bandwidth`, its quadratic term and `signal`, provides valuable insights into mobile data speed, with both the F-statistic and all coefficient t-tests being highly significant. The fitted model suggests that as available network bandwidth and mobile signal strength increases, the mobile data speed also increases. However, as the quadratic term of the available network bandwidth increases, the mobile data speed decreases. With an R^2 value of 0.8818, the model explains approximately 88.18% of the variability in mobile data speed based on these explanatory variables. This result is satisfactory. However, there are concerns with the model diagnostics. The main concern is due to the QQ plot suggesting the normality assumption was imperfectly satisfied with the slight heavy-tails of the residuals at the lower and upper tails. This finding was concluded to be

acceptable as the deviations were considered minor. In conclusion, adding polynomial terms as per suggestion was successful in enhancing the model and improving the analysis in part b.

d) We prefer **Model 4** for explaining mobile data speed on the Heriot-Watt University campus. This is due to a multitude of factors:

- This model achieved the highest R^2 value and adjusted R^2 value, hence explaining a larger portion of the variability in mobile data speed.
- This model's diagnostic plots indicate that it follows the assumptions of normality, homoscedasticity and linearity more closely than in **Model 2**. While both Models have slight deviations on the tails in the respective QQ plots, **Model 4**'s "Residuals vs Fitted" plot shows no obvious pattern, unlike in **Model 2**'s where a slight quadratic pattern can be observed.
- It can also be seen in Figure 4 that the fitted values closely align with the actual mobile data speed values if we use **Model 4** for prediction.

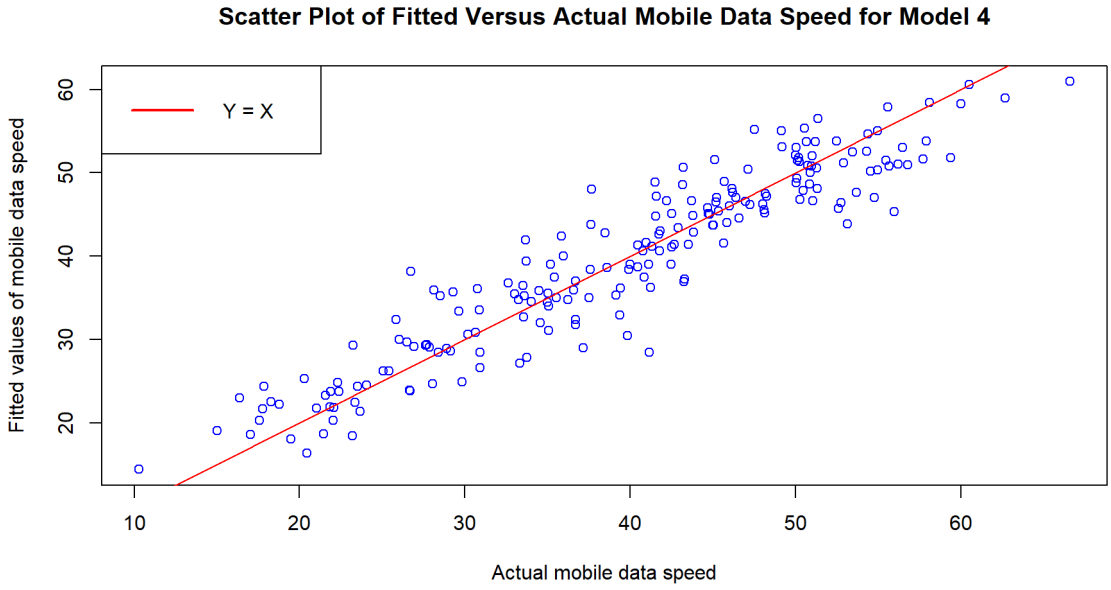


Figure 4: Scatterplot of fitted vs actual mobile data speed for **Model 4**

2. a) With a sample size of $n = 366$, collected over the year 2024, the numerical summary of the number of casual bike rentals, `casual`, is as such:

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 9.0   223.0   615.0   677.4   869.0  3065.0
```

with the sample standard deviation being 555.5068.

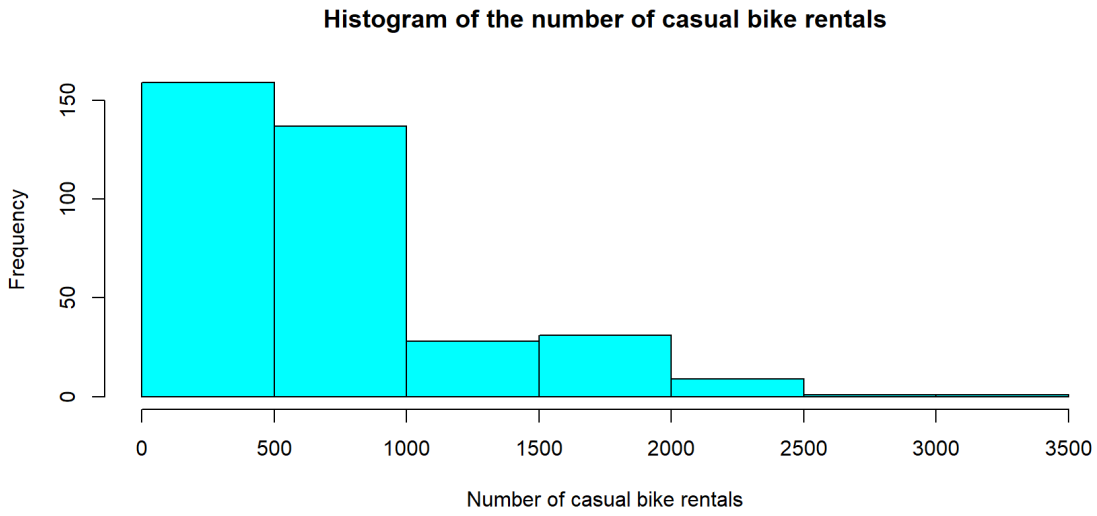


Figure 5: Histogram of the number of casual bike rentals

The histogram of the number of casual bike rentals (Figure 5) shows a unimodal, highly right-skewed distribution, which is consistent with the summary statistics, where the mean of 677.4 is significantly greater than the median of 615. The dataset ranges from a minimum bike rental of 9 to a maximum bike rental of 3065 among the 366 observations.

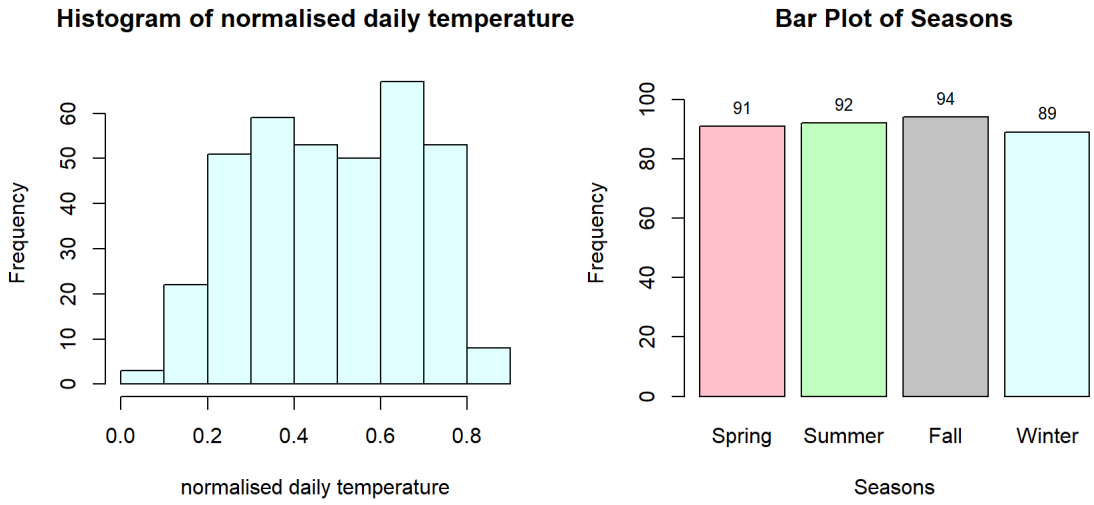


Figure 6: Distribution of data for normalised daily temperature, `temp` (left), and `seasons` (right)

The numerical summary of the normalised daily temperature, `temp`, is as such:

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.05913 0.32500 0.47750 0.48635 0.65646 0.84917
```

Sample standard deviation of the normalised daily temperature is 0.1894338.

The left figure in Figure 6 shows that the normalised daily temperature approximates a normal distribution, with a mean of 0.48635 and a standard deviation of 0.1894338. This is further

supported by both the median and mean values being quite close, suggesting a distribution resembling normality. The right figure in Figure 6 shows that 91 observations were conducted in Spring, 92 in Summer, 94 in Fall, and 89 in Winter. The distribution across the four seasons is relatively even, except for a slightly higher observation count in Fall.

b) From the left figure in Figure 7, the boxplots show the distribution of casual bike rentals for all 4 seasons (Spring, Summer, Fall, Winter). The distribution for all seasons appear to be right-skewed. Bike rentals are lowest in Spring, increase substantially in Summer, are highest in Fall, and then decline again in Winter. It is worth noting that Fall has the highest median and the widest spread, suggesting both higher typical usage and greater variability. Meanwhile, Spring has a much lower median and a tighter range, though all seasons contain some high outliers.

From the right figure in Figure 7, the scatter plot shows a generally positive relationship between temperature and casual rentals. This means that as normalised daily temperature increases, the number of casual rentals tends to rise. This pattern is strongest in the warmer seasons, especially Summer and Fall, where both temperature and rental counts are higher. On the other hand, Spring observations are concentrated at lower temperatures and lower rental counts, while Winter is more mixed but still mostly below the highest Fall values.

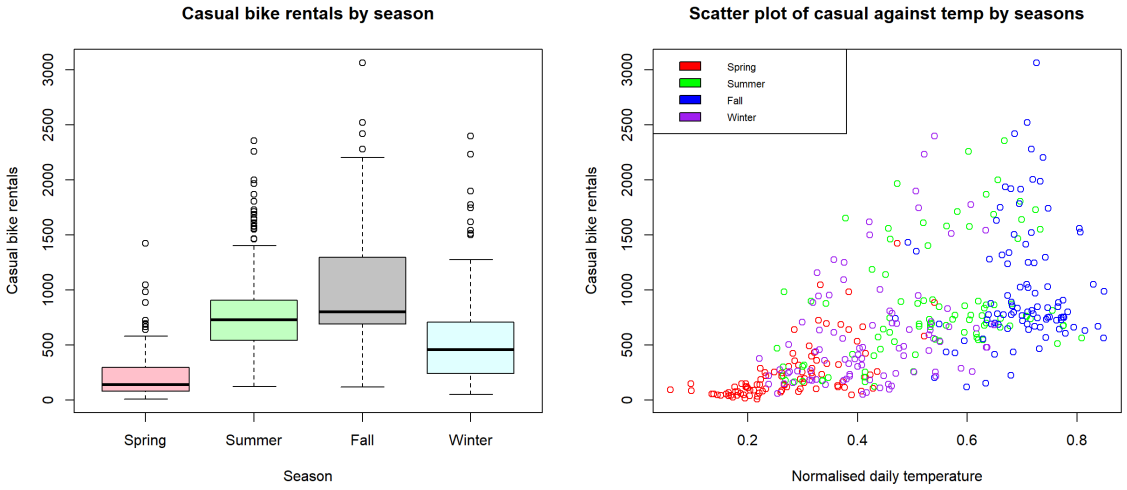


Figure 7: Boxplots of casual bike rentals (`casual`) by `season` (left), and scatterplot of casual bike rentals (`casual`) against temperature (`temp`) by `season` (right)

c) Table 1 below shows the contingency table of `holiday` status and `season` status. As observed, some cells contain very small numbers, with some having counts as low as 2. Due to limitations of conducting a χ^2 test of independence with these low counts, Fisher's exact test is used as a suitable alternative.

Table 1: Contingency table of `holiday` status and `season` status of 366 observations.

Holiday	Season			
	Spring	Summer	Fall	Winter
'No = 0'	88	90	92	86
'Yes = 1'	3	2	2	3

We use the Fisher test of independence to test:

$$H_0 : \texttt{holiday} \text{ and } \texttt{season} \text{ are independent}$$
$$H_1 : \texttt{holiday} \text{ and } \texttt{season} \text{ are not independent}$$

```
Fisher's Exact Test for Count Data
data:  matrix1
p-value = 0.8748
alternative hypothesis: two.sided
```

Based on the R output above, the associated p-value for the Fisher's exact test is 0.8748. Hence, we fail to reject H_0, concluding that `holiday` and `season` are independent at a 5% significance level.

Table 2 below shows the contingency table of `holiday` status and `workingday` status. As observed, some cells contain very small numbers, with some having counts as low as 0. Due to limitations of conducting a χ^2 test of independence with these low counts, Fisher's exact test is used as a suitable alternative.

Table 2: Contingency table of `holiday` status and `workingday` status of 366 observations.

Holiday	Working day	
	No	Yes
'No = 0'	106	250
'Yes = 1'	10	0

Again, we use the Fisher test of independence to test:

$$H_0 : \texttt{holiday} \text{ and } \texttt{workingday} \text{ are independent}$$
$$H_1 : \texttt{holiday} \text{ and } \texttt{workingday} \text{ are not independent}$$

```
Fisher's Exact Test for Count Data
data:  matrix2
p-value = 7.769e-06
alternative hypothesis: true odds ratio is not equal to 1
```

Based on the R output above, the associated p-value for the Fisher's exact test is 7.769×10^{-6}. Hence, we we have strong evidence against H_0. Thus, we reject H_0, concluding that there exists an association between `holiday` and `workingday` at a 5% significance level.

d) Fitting a Poisson GLM to model `casual`, incorporating `temp`, `season`, and their interaction (**Model A**) yields the following output:

```
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)           3.62933    0.02299  157.87   <2e-16 ***
temp                  6.11935    0.06545   93.49   <2e-16 ***
factor(season)2       2.10305    0.02787   75.45   <2e-16 ***
factor(season)3       2.84025    0.03946   71.98   <2e-16 ***
factor(season)4       1.59013    0.02978   53.40   <2e-16 ***
temp:factor(season)2 -4.29751    0.07081  -60.69   <2e-16 ***
temp:factor(season)3 -5.47416    0.07959  -68.78   <2e-16 ***
temp:factor(season)4 -3.47923    0.07686  -45.27   <2e-16 ***
```

```
Null deviance: 156292  on 365  degrees of freedom
Residual deviance:  86716  on 358  degrees of freedom
AIC: 89640
```

The individual hypothesis test indicates that all the explanatory variables, coefficients for the levels and their interactions are highly statistically significant, as the p-values are less than 2×10^{-16} for all of these t-tests. These findings suggests that all the terms significantly contribute to explaining the variation in the response variable `casual` in **Model A**. Additionally, the intercept also shows statistical significance (p-value $< 2 \times 10^{-16}$).

To further investigate the importance of the terms in the model, an Analysis of Deviance for **Model A** is performed, giving the following results:

```
                   Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                                 365     156292
temp                1    57866       364      98426 < 2.2e-16 ***
factor(season)      3     6720       361      91706 < 2.2e-16 ***
temp:factor(season) 3     4990       358      86716 < 2.2e-16 ***
```

It shows that `temp` is highly significant ($p < 2.2 \times 10^{-16}$); in the presence of `temp`, the inclusion of `season` is also significant ($p < 2.2\times10^{-16}$); in the presence of `temp` and `season`, the inclusion of the interaction between the previous 2 terms is also significant ($p < 2.2 \times 10^{-16}$). The results suggest that all terms should be retained in the model.

Next, we fit the models below separately, first without the `temp` term, then without the `season` term, and so on, along with the null model. The following deviance residuals and AIC values were obtained.

Model	**Terms present**	**Deviance**	**df**	**AIC**
Model A	temp + season + temp:season	86715.94	358	89640.45
Model B	temp + temp:season	94561.49	361	97479.99
Model C	season + temp:season	86715.94	358	89640.45
Model D	temp + season	91705.72	361	94624.22
Model E	Null	156291.70	365	159202.20

Analysis of Deviance:

Comparing **Model A** with **Model B**, change in deviance is $94561.49 - 86715.94 = 7845.55$ on 3 df, with p-value, $p = P(\chi^2_3 > 7845.55) < 2.2 \times 10^{-16}$. Removing the `season` term from the model does significantly reduce goodness-of-fits.

It follows that **Model A** and **Model C** are both the same model essentially, even though `season` term has been removed. Hence, the 2 models are not comparable.

Comparing **Model A** with **Model D**, change in deviance is $91705.72 - 86715.94 = 4989.78$ on 3 df, with p-value, $p = P(\chi^2_3 > 4989.78) < 2.2 \times 10^{-16}$. Removing the interaction term from the model does significantly reduce goodness-of-fits.

From analysis of deviance results, all terms should be retained in the model. Thus, we prefer **Model A**.

Comparing **Model A** with the null model, change in deviance is $156291.70-86715.94 = 69575.76$ on 7 df, with p-value, $p = P(\chi^2_7 > 69575.76) < 2.2 \times 10^{-16}$. So, strong evidence that **Model A** fits better than the null model.

Comparing **Model A** with the saturated model, the residual deviance is 86715.94 on 358 df, with the associated p-value, $p = P(\chi^2_3 58 > 86715.94) = 0$. Overwhelming evidence against goodness-of-fit of the model. That is, the **Model A** with all terms included is significantly worse than the saturated model.

Looking at the AIC values, **Model A** has the lowest AIC of 89640.45. The differences in AIC between **Model A** and other models are as follows: 7839.54 for **Model B** ($97479.99 - 89640.45$), and 4983.77 for **Model D** ($94624.22 - 89640.45$). Since these differences exceed the rule-of-thumb threshold of 2, this suggests that **Model A** is the best model overall.

Hence, **Model A** with explanatory variables `temp`, `season` and their interaction included is preferred. The fitted model is:

$$\begin{aligned}\log(\mu_i) = {} & 3.62933 + 6.11935\,(\text{temp}_i) + 2.10305 I(\text{season}_i = 2) + 2.84025 I(\text{season}_i = 3) \\ & + 1.59013 I(\text{season}_i = 4) - 4.29751\,(\text{temp}_i) I(\text{season}_i = 2) \\ & - 5.47416\,(\text{temp}_i) I(\text{season}_i = 3) - 3.47923\,(\text{temp}_i) I(\text{season}_i = 4)\end{aligned}$$

where μ_i is the expected number of casual rentals for day i, and $I()$ is the indicator function of `season` with Spring as the base level.

The fitted **Model A** shows that, holding `temp` constant:

• The expected number of `casual` rentals in Summer (season 2) relative to Spring is $e^{2.10305} = 8.19$, meaning that the expected number of rentals is about 8.19 times higher than in Spring.

• The expected number of `casual` rentals in Fall (season 3) relative to Spring is $e^{2.84025} = 17.12$, meaning that the expected number of rentals is about 17.12 times higher than in Spring.

• The expected number of `casual` rentals in Winter (season 4) relative to Spring is $e^{1.59013} = 4.90$, meaning that the expected number of rentals is about 4.90 times higher than in Spring.

The coefficient for `temp` indicates that, in Spring (the baseline season), a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.11935} = 454.57$, indicating a very strong positive effect of temperature.

The interaction terms show that the effect of `temp` differs by season:

• In Summer, a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.11935-4.29751} = e^{1.82184} = 6.18$.

• In Fall, a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.11935-5.47416} = e^{0.64519} = 1.91$.

• In Winter, a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.11935-3.47923} = e^{2.64012} = 14.01$.

Overall, this indicates that while higher `temp` increases the expected number of `casual` rentals in all seasons, the magnitude of this effect is strongest in Spring and Winter, and weaker in Summer and Fall.

The plot of deviance residuals against fitted values (Figure 8) shows that the residuals are not randomly scattered around zero. Firstly, the spread of the residuals appears to increase for larger fitted values, in a fan-shaped pattern, indicating potential overdispersion or non-constant variance, which violates the Poisson assumption that the variance equals the mean. Moreover, there are several large positive and negative residuals, suggesting the presence of outliers or observations that are poorly fitted by the model. These features indicate that the model may not be fully adequate.

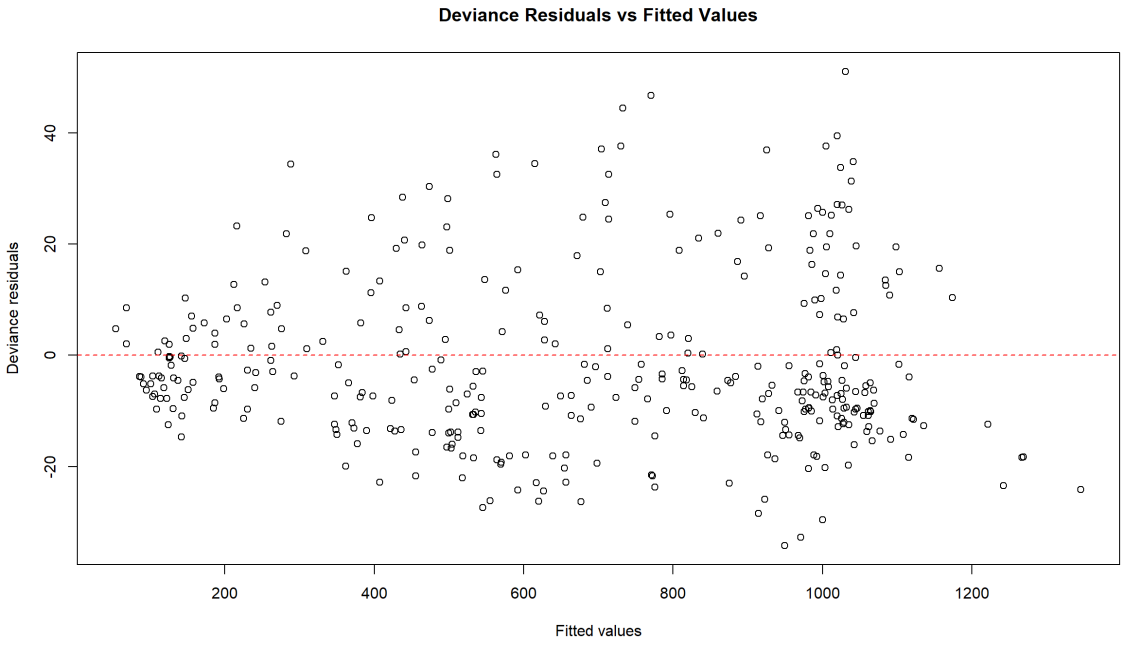


Figure 8: Deviance residuals against fitted values for **Model A**

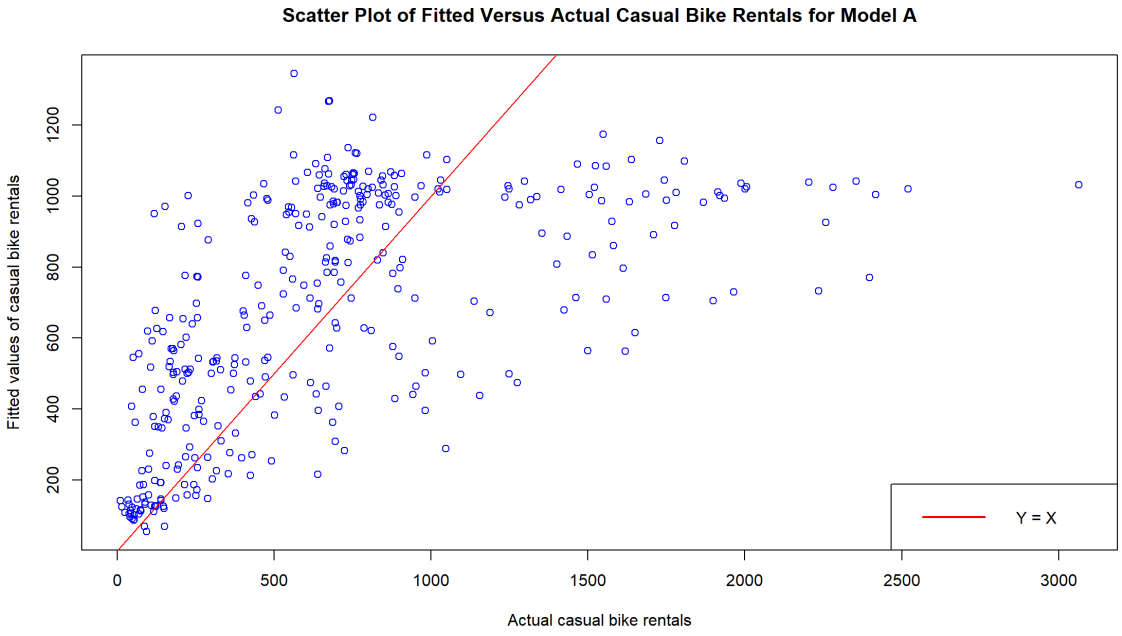


Figure 9: Scatter plot of fitted versus actual casual bike rentals in **Model A**

Conclusion: The analysis of deviance and relatively low AIC value (89640.45), with differences of AIC exceeding the threshold of 2, confirms the superiority of **Model A** amongst the models compared. However, the residual deviance plot for **Model A** suggests a major violation of model assumptions, rendering the model inadequate. It is also worth noting that **Model A** is significantly worse than the saturated model, despite being the best model overall. Observing

the results in Figure 9 further proves the point that **Model A**, while being the preferred model, is inadequate in modeling the dataset. In particular, the presence of patterns and increasing variability suggests that a more flexible model or an alternative distribution may provide a better fit. This motivates us to explore the Negative Binomial GLM in the next section.

e) Fitting a Negative Binomial GLM to model `casual`, incorporating `temp`, `season`, and their interaction (**Model F**) yields the following output:

```
                    Estimate Std. Error z value Pr(>|z|)
(Intercept)           3.3976     0.2004  16.954  < 2e-16 ***
temp                  6.8683     0.6870   9.997  < 2e-16 ***
factor(season)2       2.1581     0.3270   6.599 4.13e-11 ***
factor(season)3       3.0058     0.6570   4.575 4.77e-06 ***
factor(season)4       1.7191     0.3378   5.090 3.59e-07 ***
temp:factor(season)2 -4.7264     0.8312  -5.686 1.30e-08 ***
temp:factor(season)3 -6.1290     1.1224  -5.460 4.75e-08 ***
temp:factor(season)4 -3.9957     0.9242  -4.324 1.53e-05 ***

Null deviance: 763.51  on 365  degrees of freedom
Residual deviance: 389.49  on 358  degrees of freedom
AIC: 5234.2
```

The individual hypothesis test indicates that all the explanatory variables, coefficients for the levels and their interactions are highly statistically significant, as the p-values are less than 0.05 for all of these t-tests. These findings suggests that all the terms significantly contribute to explaining the variation in the response variable `casual` in **Model F** at a 5% confidence level. Additionally, the intercept also shows statistical significance (p-value $< 2 \times 10^{-16}$).

The adequacy of the fitted **Model F** was assessed by comparing it to the saturated model via the residual deviance. The ratio of residual deviance to degrees of freedom is 1.087953, which is close to 1 (the dispersion parameter) and suggests that **Model F** provides a good fit to the data with no substantial lack of fit.

Looking at the AIC value, **Model F** achieves an AIC value of 5234.2. This value is much lower than the previous **Model A**, indicating model adequacy.

The fitted model is:

$$\begin{aligned}\log(\mu_i) = {} & 3.3976 + 6.8683 \cdot \text{temp}_i + 2.1581 \cdot \mathbf{1}(\text{season}_i = 2) + 3.0058 \cdot \mathbf{1}(\text{season}_i = 3) \\ & + 1.7191 \cdot \mathbf{1}(\text{season}_i = 4) - 4.7264 \cdot \text{temp}_i \cdot \mathbf{1}(\text{season}_i = 2) \\ & - 6.1290 \cdot \text{temp}_i \cdot \mathbf{1}(\text{season}_i = 3) - 3.9957 \cdot \text{temp}_i \cdot \mathbf{1}(\text{season}_i = 4)\end{aligned}$$

where μ_i is the expected number of casual rentals for day i, and $I()$ is the indicator function of `season` with Spring as the base level.

The fitted **Model F** shows that, holding `temp` constant:

• The expected number of `casual` rentals in Summer (season 2) relative to Spring is $e^{2.1581} = 8.65$, meaning that the expected number of rentals is about 8.65 times higher than in Spring.

• The expected number of `casual` rentals in Fall (season 3) relative to Spring is $e^{3.0058} = 20.20$, meaning that the expected number of rentals is about 20.20 times higher than in Spring.

• The expected number of `casual` rentals in Winter (season 4) relative to Spring is $e^{1.7191} = 5.58$, meaning that the expected number of rentals is about 5.58 times higher than in Spring.

The coefficient for `temp` indicates that, in Spring (the baseline season), a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.8683} = 961.31$, indicating a very strong positive effect of temperature.

The interaction terms show that the effect of `temp` differs by season:

• In Summer, a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.8683-4.7264} = e^{2.1419} = 8.52$.

• In Fall, a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.8683-6.1290} = e^{0.7393} = 2.09$.

• In Winter, a one-unit increase in `temp` multiplies the expected number of `casual` rentals by $e^{6.8683-3.9957} = e^{2.8726} = 17.68$.

Overall, this indicates that while higher `temp` increases the expected number of `casual` rentals in all seasons, the magnitude of this effect is strongest in Spring and Winter, and weaker in Summer and Fall.

The plot of deviance residuals against fitted values (Figure 10) shows that the residuals are randomly scattered around zero. Moreover, the residuals are small, suggesting no outliers or observations that are poorly fitted by the model. These features show that the model is performing adequately.

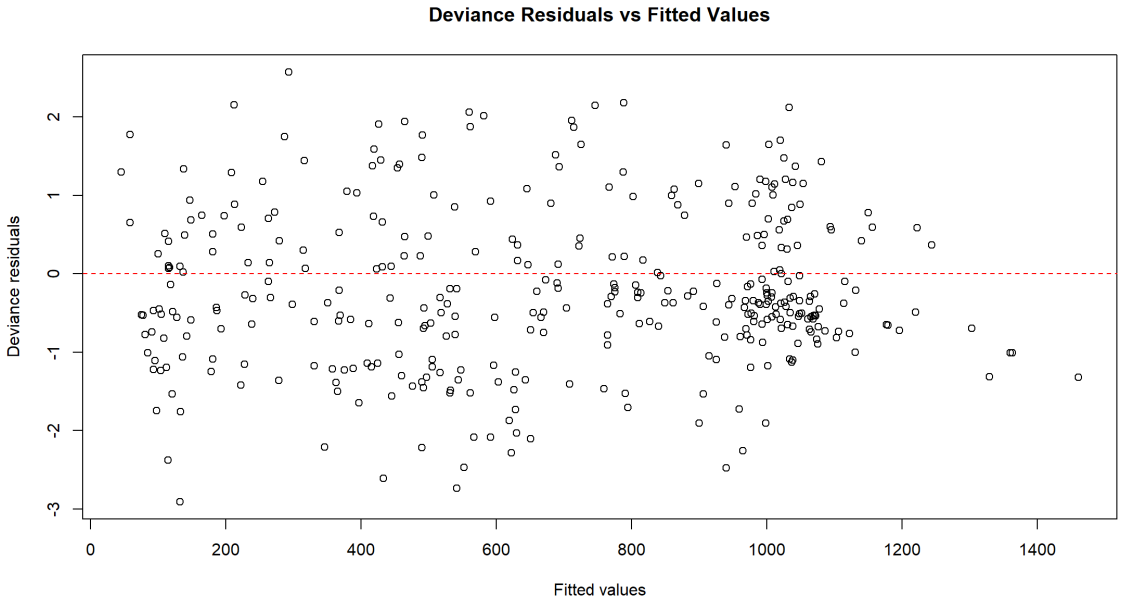


Figure 10: Deviance residuals against fitted values for **Model F**

Conclusion: The results from **Model F** can be compared directly with those obtained from **Model A**. Although both models share the same mean structure and include identical explanatory variables and interaction terms, their adequacy differs substantially due to the underlying distributional assumptions.

From the diagnostic analysis of **Model A** (Figure 8), there is clear evidence of overdispersion, as indicated by the fan-shaped pattern and increasing spread of residuals with fitted values. This violates the Poisson assumption that the variance equals the mean, leading to an inadequate model fit despite strong statistical significance and a relatively low AIC within the class of Poisson

models. Furthermore, Figure 9 shows minimal alignment in the fitted values against actual values across temperature and season, reinforcing the inadequacy of **Model A** in capturing the true structure of the data.

In contrast, **Model F** addresses this issue by incorporating an additional dispersion parameter through the Negative Binomial distribution. As observed in Figure 10, the deviance residuals are randomly scattered around zero with no discernible pattern, indicating that the issue of overdispersion has been effectively resolved. This provides strong evidence that **Model F** is more appropriate for the data.

In addition, **Model F** achieves a substantially lower AIC value (5234.2) compared to **Model A** (89640.45). This large reduction in AIC indicates a significantly improved model fit, even after accounting for model complexity. Combined with the deviance-to-degrees-of-freedom ratio being close to 1, this confirms that **Model F** provides an adequate fit to the data.

Overall, while **Model A** was the preferred model within the class of Poisson GLMs, it is inadequate due to violations of model assumptions, particularly overdispersion. **Model F**, on the other hand, successfully overcomes these limitations and demonstrates superior performance in terms of goodness-of-fit, diagnostics, and model selection criteria.

Therefore, **Model F** is the preferred model for this analysis.

Finally, it would be wise to consider including `workingday` as an additional explanatory variable into the model. This is because knowing the `workingday` status would allow the model to predict the extreme tails with more confidence and certainty, as people tend to have more free time during holidays or non-working days.

f) It is not appropriate to include both `holiday` and `workingday` as explanatory variables in the same count regression model, whether Poisson or Negative Binomial, because they are structurally dependent and therefore do not provide independent information.

Using the observed contingency table previously constructed (Table 2), we see that all observations with `holiday = 1` occur when `workingday = 0`. Thus, in this dataset, holiday status is effectively embedded within working-day status. This creates substantial overlap between the two predictors.

Testing Fisher's exact test gives a p-value of 7.769×10^{-6}, obtained from previous analysis in part c.

Hence, there is very strong evidence against independence at the 5% significance level, indicating that `holiday` and `workingday` are strongly associated.

Therefore, including both variables in the same model would introduce redundant information and may cause instability or poor interpretability of the estimated coefficients. Since this is a problem with the explanatory variables rather than the response distribution, the same reasoning applies to both Poisson and Negative Binomial models. For this reason, only one of the two variables should be included in the final model.

A. Appendix

```r
# Loading library required
library(MASS)

################################################################################
# Q1
################################################################################

# Loading the data frame and attaching it
filepath = "C:/Users/marcu/OneDrive/Desktop/02_Courses/03_Stat Models B/04_Assignment/Assignment 2/datasets/MobileNetwork.txt"
mobile_network = read.table(filepath,
                    header = TRUE)

attach(mobile_network)

# Part A
# Function to calculate and display correlations (to be looped)
panel.cor <- function(x, y, ...) {
  usr <- par("usr"); on.exit(par(usr)) # retrieve coordinates of each plot
  par(usr = c(0, 1, 0, 1)) # set x and y axis range from 0 to 1
  r <- round(cor(x, y), 2) # calculate correlations
  text(0.5, 0.5, r, cex = 1.5) # place result in middle (0.5,0.5)
}

# Pairwise plot
pairs(mobile_network,
      lower.panel = panel.smooth, # Trend line
      upper.panel = panel.cor, # Correlation panels
      main = "Pairwise Relationships Between Variables")

# Part B
# Model 1
lm.model1 = lm(speed ~ bandwidth + signal + legacy_users)
summary(lm.model1)
summary.aov(lm.model1)

# Model 2 (Preferred model)
lm.model2 = lm(speed ~ bandwidth + signal)
summary(lm.model2)
summary.aov(lm.model2)

# Diagnostic plots for preferred model
par(mfrow=c(2,2))
plot(lm.model2, which = 1:4)

# Part C
# Model 3
lm.model3 = lm(speed ~ bandwidth +  signal * I(bandwidth^2))
summary(lm.model3)
summary.aov(lm.model3)

# Model 4 (Preferred model)
lm.model4 = lm(speed ~ bandwidth + signal + I(bandwidth^2))
summary(lm.model4)
summary.aov(lm.model4)

# Diagnostic plots for preferred model
plot(lm.model4, which = 1:4)
cooks.distance(lm.model4)[186]

# Part D
# Comparison between best models from parts b and c
# Plot of fitted values against actual values for Model 4
par(mfrow=c(1,1))

plot(speed, lm.model4$fitted.values,
     xlab = "Actual mobile data speed",
     ylab = "Fitted values of mobile data speed",
     main = "Scatter Plot of Fitted Versus Actual Mobile Data Speed for Model 4",
     col = "blue")
```

```r
abline(a = 0, b = 1, col = "red")
legend("topleft",
       legend = "Y = X",
       col = "red",
       lty = 1,
       lwd = 2)

# Detaching dataset
detach(mobile_network)

################################################################################
# Q2
################################################################################

# Loading the data frame and attaching it
rent <- read.table("rent.csv",
                   header = TRUE, sep = ",")

attach(rent)

# Part A
# Numerical summaries for:
# casual
summary(casual)
sd(casual)
hist(casual,
     main = "Histogram of the number of casual bike rentals",
     xlab = "Number of casual bike rentals",
     col = "cyan")

# temp
par(mfrow=c(1,2))
summary(temp)
sd(temp)
hist(temp,
     main = "Histogram of normalised daily temperature",
     xlab = "normalised daily temperature",
     col = "lightcyan")

# season
season_freq = table(season)

bar = barplot(season_freq,
        xlab = "Seasons",
        ylab = "Frequency",
        main = "Bar Plot of Seasons",
        names.arg = c("Spring", "Summer", "Fall", "Winter"),
        ylim = c(0,110),
        col = c("pink", "darkseagreen1", "gray76", "lightcyan"),
        border = "black")

text(x= bar, y=season_freq, pos = 3, label = season_freq, cex = 0.8)

# Part B
# Plot of casual against season
par(mfrow=c(1,2))
season_cleaned = factor(season)

boxplot(casual ~ season,
        xlab = "Season", ylab = "Casual bike rentals",
        names = c("Spring", "Summer", "Fall", "Winter"),
        main = "Casual bike rentals by season",
        col = c("pink", "darkseagreen1", "gray76", "lightcyan"))

# Plot of casual against temp, distinguished by season
plot(casual ~ temp, type='n',
     main = "Scatter plot of casual against temp by seasons",
     xlab = "Normalised daily temperature",
     ylab = "Casual bike rentals")
points(casual[season == 1] ~ temp[season == 1], col = "red")
```

```r
points(casual[season == 2] ~ temp[season == 2], col = "green")
points(casual[season == 3] ~ temp[season == 3], col = "blue")
points(casual[season == 4] ~ temp[season == 4], col = "purple")
legend("topleft", legend=c("Spring", "Summer", "Fall", "Winter"),
       fill=c("red", "green", "blue", "purple"),
       cex = 0.7)

# Part C
# Creating contingency matrices for both cases
# Case 1
matrix1 = matrix(c(sum(season == 1 & holiday == 0),
                   sum(season == 1 & holiday == 1),
                   sum(season == 2 & holiday == 0),
                   sum(season == 2 & holiday == 1),
                   sum(season == 3 & holiday == 0),
                   sum(season == 3 & holiday == 1),
                   sum(season == 4 & holiday == 0),
                   sum(season == 4 & holiday == 1)), 2, 4)
matrix1

# Case 2
matrix2 = matrix(c(sum(workingday == 0 & holiday == 0),
                   sum(workingday == 0 & holiday == 1),
                   sum(workingday == 1 & holiday == 0),
                   sum(workingday == 1 & holiday == 1)), 2, 2)
matrix2

# Since both matrices have expected frequency cells lesser than 5,
# we use Fisher's exact test.
fisher.test(matrix1)
fisher.test(matrix2)

# Part D
# Main model
glm.model1 = glm(casual ~ temp * factor(season), family = poisson)
summary(glm.model1)
anova(glm.model1)

# Nested models
glm.model2 = glm(casual ~ temp + temp : factor(season), family = poisson)
glm.model3 = glm(casual ~ factor(season) + temp : factor(season), family = poisson)
glm.model4 = glm(casual ~ temp + factor(season), family = poisson)
glm.model5 <- glm(casual ~ 1, family = poisson)

# Making all models into a list for easier looping.
glm.models = list(glm.model1 = glm.model1,
                  glm.model2 = glm.model2,
                  glm.model3 = glm.model3,
                  glm.model4 = glm.model4,
                  glm.model5 = glm.model5)

# Residual Deviance for each model
res_dev_glm = sapply(glm.models, deviance)
res_dev_glm

# Residual Deviance's degrees of freedom for each model
res_dev_df_glm = sapply(glm.models, df.residual)
res_dev_df_glm

# AIC values for each model
AIC_glm = sapply(glm.models, AIC)
AIC_glm

# Analysis of Deviance
anova(glm.model2, glm.model1)
anova(glm.model3, glm.model1) # Note: Essentially the same model
anova(glm.model4, glm.model1)
anova(glm.model5, glm.model1)

# p-value to test significance of nb.model against maximal model
1 - pchisq(glm.model1$deviance, glm.model1$df.residual)
```

```r
# Plot of deviance residuals against fitted values
par(mfrow=c(1,1))
plot(fitted(glm.model1),
     residuals(glm.model1, type = "deviance"),
     xlab = "Fitted values",
     ylab = "Deviance residuals",
     main = "Deviance Residuals vs Fitted Values"
)

abline(h = 0, lty = 2, col = "red")

# Plot of fitted values against actual values
plot(casual, glm.model1$fitted.values,
     xlab = "Actual casual bike rentals",
     ylab = "Fitted values of casual bike rentals",
     main = "Scatter Plot of Fitted Versus Actual Casual Bike Rentals for Model A",
     col = "blue")

abline(a = 0, b = 1, col = "red")
legend("bottomright",
       legend = "Y = X",
       col = "red",
       lty = 1,
       lwd = 2)

# Conclusion: Choose glm.model1

# Part E
# Fitting same model into negative binomial model
nb.model = glm.nb(casual ~ temp * factor(season))
summary(nb.model)

# Compare against saturated model
nb.model$deviance / nb.model$df.residual

# Plot of deviance residuals against fitted values
plot(fitted(nb.model),
     residuals(nb.model, type = "deviance"),
     xlab = "Fitted values",
     ylab = "Deviance residuals",
     main = "Deviance Residuals vs Fitted Values"
)

abline(h = 0, lty = 2, col = "red")

# Conclusion, nb.model more adequate
# Part F to be explained in the report.
```